

AD23633-GENERATIVE AI

- **Problem Statement Title**-AI Chatbot & Virtual Front Desk for Government Hospitals with Health Awareness Integration
- **Theme**-MedTech / BioTech / HealthTech
- **PS Category**- Software
- **Team Name**- InnoWave
- **Team Members**-Gayathri R(231801039)



AI-Driven Public Health Chatbot for Disease Awareness

Detailed Solution

- Provides healthcare access via **app, WhatsApp, SMS, and calls.**
- Offers **symptom prediction, report analysis, remedies, reminders, and alerts.**
- Escalates complex cases to doctors for **verified guidance.**

Problem Addressed

- **Low healthcare awareness** in rural and semi-urban areas.
- Addresses **language barriers.**
- Limited access to doctors and **reliable health advice.**

Innovation & Uniqueness

- **AI + NLP** with regional language voice recognition.
- Human-in-the-loop **verification for complex cases.**
- Unique blend of **remedies, preventive care, and voice alerts** in one system.

Problem statement

Problem Statement: Public Health Awareness Gap

- A large population in **rural and semi-urban areas** lacks **timely and reliable health information**.
- **Language barriers**, low digital literacy, and **poor internet connectivity** prevent people from accessing accurate medical guidance.
- **Shortage of doctors** and long travel distances lead to delayed consultations and untreated conditions.
- Many people depend on **unverified sources**, home guesses, or delay care until conditions worsen.
- Existing digital health solutions are often **complex, English-centric, and not inclusive** for elderly or less-educated users.

Problem Statement

Public Health Awareness Gap

Problem Issues:

-  Low health awareness in rural & semi-urban areas
-  Language barriers & poor internet connectivity
-  Shortage of doctors & distant healthcare access
-  Reliance on unverified advice & home remedies

Resulting Impact:

-  Late disease detection & untreated conditions
-  Overcrowded hospitals for minor ailments
-  Higher costs & health risks



The need for an **accessible, multilingual, and trusted health guidance system.**

Sustainable Development Goals (SDGs)

📌 **SDG 2 – Zero Hunger**

NutriScan AI supports improved nutrition by providing personalized dietary analysis, enabling individuals to make healthier food choices and helping identify and reduce malnutrition risks.

📌 **SDG 3 – Good Health and Well-Being**

By analyzing food safety and nutritional content, the system helps prevent food-borne illnesses, supports balanced diets, and promotes overall health and well-being.

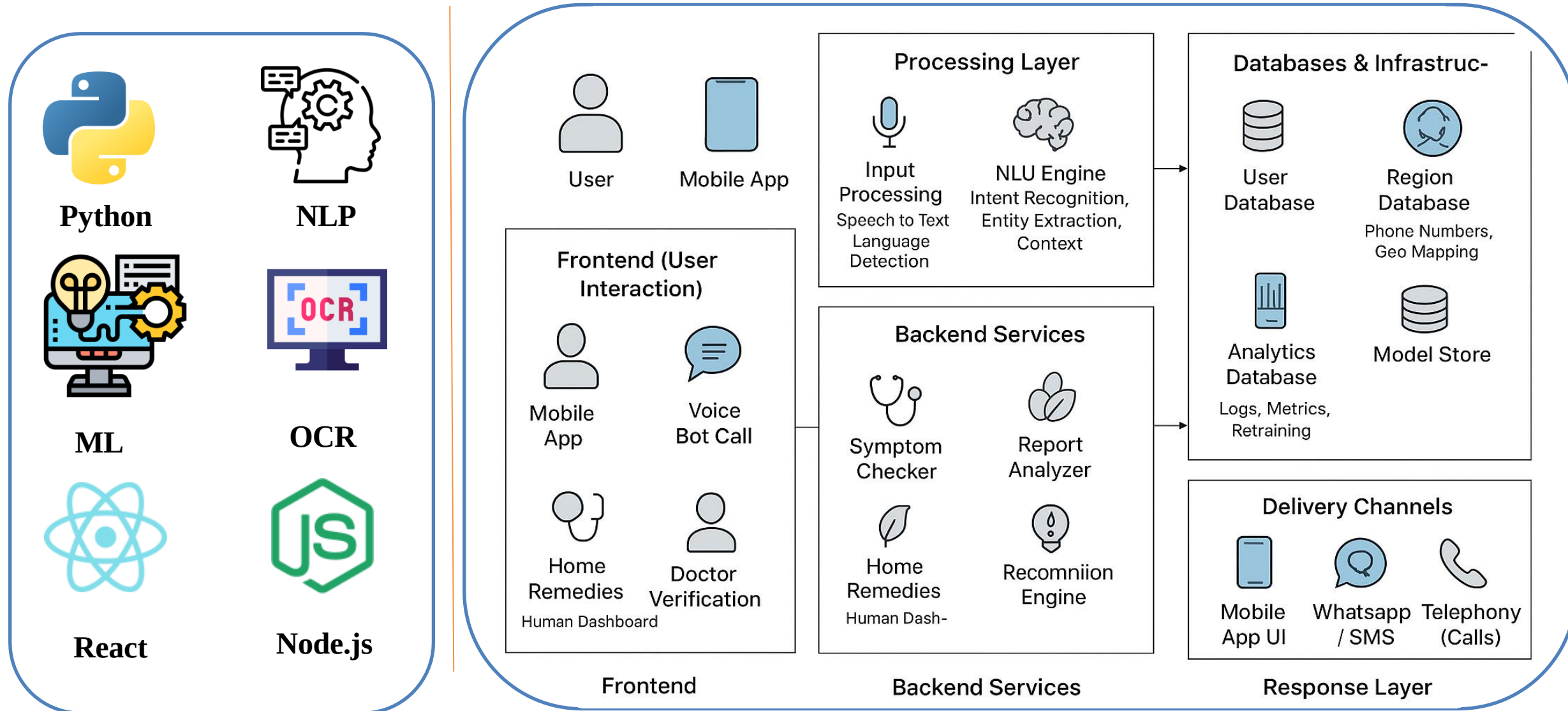
📌 **SDG 9 – Industry, Innovation, and Infrastructure**

The project leverages AI agents and intelligent data analysis to drive innovation in digital health, food technology, and smart healthcare infrastructure.

📌 **SDG 12 – Responsible Consumption and Production**

NutriScan AI encourages responsible food consumption by promoting informed eating habits, reducing food waste, and increasing awareness of food quality and safety.

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

Accessible

- Support via App, WhatsApp, SMS, and Voice calls.
- Voice in regional languages ensures inclusivity

Usability

- Simple chatbot flow with minimal typing.
- Icon-based UI for easy navigation.

Trust

- AI responses verified by doctors.
- Data connected with govt health alerts

Economic

- Free or low-cost service for users.
- SMS/voice services can be funded by NGOs/government.

Challenges

- Language accuracy across multiple regional dialects.
- Poor internet connectivity in rural areas.
- Data privacy and user trust issues.

Solutions

- Fine-tuned AI/NLP models with multilingual training.
- Offline and SMS-based fallback channels.
- End-to-end encryption for data security..

Viability

- **Adaptability** – Regular updates and integration with health alerts keep the system useful.
- **Healthcare Impact** – Improves preventive care, reduces hospital congestion, and bridges rural–urban gaps.
- **Reliability** – Doctor verification and government-linked data strengthen credibility.

IMPACT AND BENEFITS



Government Doctors Online

Impact: Government doctors become accessible online for consultations.

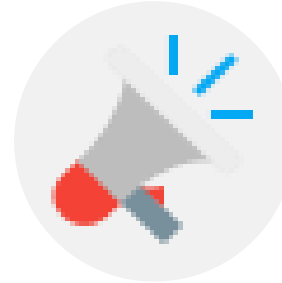
Benefit: People consult them without stigma of “low priority” or “unclean” facilities, improving trust in public healthcare.



Voice Access in Local Language

Impact: Elderly and less literate users access health info through voice & video calls.

Benefit: Ensures inclusivity, better awareness, and wider adoption.



Regional Health Alert

Impact: Villagers receive outbreak alerts (like dengue or malaria).

Benefit: Communities take precautions early, reducing disease spread.



Costly Hospital Visit Avoided

Impact : Chatbot suggests safe home remedies or guides when hospital visit is not required.

Benefit: Saves money and reduces unnecessary hospital crowding.

RESEARCH AND REFERENCES

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